

Anticoagulation Management

Policy Number	POL-PH-030	Owner / Department	Pharmacy / Nursing
Effective Date	2023-06-01	Version	3.1
Last Revised	2026-03-12	Next Review	2028-03-12
Approved By	Pharmacy & Therapeutics Committee	Applies To	All prescribers, pharmacists, and nurses who order, dispense, administer, or monitor therapeutic anticoagulation in any inpatient or observation unit

1. Purpose

To reduce patient harm associated with therapeutic anticoagulant therapy through standardized ordering, weight-based dosing, laboratory monitoring, patient education, and defined reversal protocols, in compliance with The Joint Commission National Patient Safety Goal NPSG.03.05.01.

2. Scope

Applies to all therapeutic anticoagulation: warfarin, unfractionated heparin (UFH) infusions, low-molecular-weight heparins (LMWH), and direct oral anticoagulants (DOACs). It does not apply to routine short-term prophylactic venous-thromboembolism (VTE) prevention related to a procedure or hospitalization, which is addressed by the VTE prophylaxis order set.

3. Definitions

- **Anticoagulant** — a medication that reduces the blood's ability to clot; a high-alert medication per the ISMP list.
- **DOAC** — direct oral anticoagulant (e.g., apixaban, rivaroxaban, dabigatran, edoxaban).
- **Therapeutic range** — the target INR band for warfarin (commonly 2.0–3.0; 2.5–3.5 for mechanical mitral valves) or the aPTT/anti-Xa band for a heparin infusion.
- **Reversal agent** — a product that restores hemostasis by countering an anticoagulant (e.g., vitamin K, 4-factor PCC, idarucizumab, andexanet alfa, protamine sulfate).

4. Policy and Procedure

Use the approved weight-based order sets and nomograms for all anticoagulant orders; free-text anticoagulant orders are not permitted. Current weight in kilograms must be in the record before the first dose. Baseline labs (CBC, INR/PT, aPTT, renal function) are obtained before initiation and monitored per the nomogram.

1. Verify indication, current weight (kg), renal/hepatic function, and baseline coagulation labs before the first dose.
2. Enter the order through the approved anticoagulant order set; pharmacy performs an independent dose and interaction check before dispensing.

- Two nurses independently verify drug, concentration, pump settings, and dose for all heparin infusions (independent double check).
- Monitor the appropriate parameter (INR, aPTT, or anti-Xa) on the schedule defined by the nomogram and adjust per protocol.
- Provide anticoagulant-specific education to the patient and family and document understanding before discharge.

IMPORTANT — For any major or life-threatening bleed on an anticoagulant, hold all further doses immediately, notify the prescriber STAT, and initiate the approved reversal protocol for that specific agent without waiting for lab results. Treat as SEV-1 and notify Risk Management within 1 hour.

5. Reversal Agents by Anticoagulant

The following approved reversal strategies are stocked and available on a STAT basis.

Anticoagulant	First-line reversal	Notes
Warfarin	Vitamin K + 4-factor PCC	Dose PCC by weight & INR; give FFP if PCC unavailable.
UFH / LMWH	Protamine sulfate	Weight- & time-based dosing; partial LMWH reversal only.
Dabigatran	Idarucizumab	Dialyzable; idarucizumab is the specific antidote.
Apixaban / rivaroxaban	Andexanet alfa	4-factor PCC is an alternative if unavailable.

6. Severity & Reporting

Classify anticoagulation events on the facility-wide scale in POL-RM-013 and report through the event system; medication errors are additionally graded by the NCC MERP A–I harm index per POL-PS-003.

Severity	Example	Notify / timeframe
SEV-1	Fatal or life-threatening bleed; intracranial hemorrhage requiring reversal.	Prescriber, Nursing Supervisor & Risk Mgmt within 1 hour.
SEV-2	Supratherapeutic INR with bleed requiring transfusion or prolonging stay.	Prescriber & Risk Mgmt within 4 hours.
SEV-3 / Near Miss	Missed INR draw with no harm; wrong dose intercepted by pharmacy.	Event report within 24 hours.

7. Documentation Requirements

- Indication, agent, dose, route, and current weight (kg) used for dosing.
- Baseline and ongoing monitoring values (INR, aPTT, or anti-Xa) against the target range.
- Independent double check performed for heparin infusions, with both verifier names.
- Anticoagulant-specific patient/family education and confirmation of understanding.

- Any reversal agent given, with dose, time, and prescriber notification.

8. References

- The Joint Commission NPSG.03.05.01 — Reduce the likelihood of patient harm associated with the use of anticoagulant therapy.
- ISMP List of High-Alert Medications in Acute Care Settings.
- CMS Conditions of Participation, 42 CFR 482.25 (pharmaceutical services).
- Related: POL-RM-013 (severity & reporting); POL-PS-003 (medication-error reporting); POL-PH-032 (high-alert medications).